K-pop PageRank Report

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The popularity of K-pop is steadily growing worldwide. In this context, I have long held a particular notion: that artists affiliated with major agencies—those generating massive revenue through high sales figures year after year—owe their success largely to the significant influence and backing provided by their respective agencies. In Korea, HYBE, SM, JYP, and YG are recognized as the leading entertainment agencies for idol groups. Among these, HYBE stands out as the premier agency, having acquired numerous mid-sized companies that house popular idol groups, and it consistently records the highest revenue figures in the industry. Consequently, I investigated the mutual influence exerted between 288 male and female idol groups and their respective agencies, aiming to identify which agencies and idol groups possess the highest levels of influence. Furthermore, by analyzing data from four agencies with particularly high revenue figures, I examined the extent to which this factor impacts the idol groups under their management.

To generate the results for this study, I utilized graph data and the PageRank algorithm. The process consisted of four distinct stages; the first stage involved data collection and data cleaning. To begin with, relying on a single dataset proved insufficient to uncover the results I was seeking; therefore, I utilized a total of three distinct datasets. I then leveraged these to construct a new dataset.

The first dataset contained information on a total of 1,778 K-pop idols; from this pool, I decided to focus exclusively on group artists. This naturally raises the question: Are there not also successful solo artists? However, the distinction lies in the fact that "singers" and "idols" differ in both their conceptual definitions and their career trajectories. Idols are typically subject to a greater degree of influence from their management agencies, and they generally operate as members of groups rather than as solo acts. Idol groups are entities built around planning, concepts, and branding, whereas a singer focuses purely on their own vocal prowess. For instance, PSY—who reached number two on the Billboard charts in 2012 with the hit song "Gangnam Style"—is classified as a "singer," not an "idol." Upon filtering the dataset to include only idol groups, I identified a total of 288 entries, which also included the agencies to which they belong.

For the second dataset, I utilized the activity status of idol groups, music awards, end-of-year awards, physical sales, and YouTube view counts. While collecting data, I discovered that even idol groups signed to major entertainment agencies struggle to sustain their success. According to the dataset, a significant 24 percent of 120 groups were either currently on hiatus or had already disbanded. Although there are various reasons for this, the primary factor is that—despite idol groups generally boasting high success rates—most ultimately fail to renew their contracts when the term expires, or the team disbands due to personal circumstances.

In the third dataset, we included follower counts and the current popularity rankings of the groups. Next, I performed data cleaning. Since I was working with multiple datasets, I identified and resolved any mismatches in the information and filled in the missing values with zeros. Furthermore, to analyze the influence of major entertainment agencies, the actual revenue figures for HYBE, SM, JYP, and YG for the year 2024 were subsequently included.

[118]:

	Group	Company
0	7 O'clock	Jungle
1	Limitless	ONO
2	VAV	A team
3	Hash Tag	LUK
4	MOMOLAND	Double Kick
...	...	...
1211	SNUPER	Widmay
1293	UNIQ	Yuehua
1308	G-reyish	Elijah
1373	Gfriend	Source
1384	Luminous	Barunson

288 rows × 2 columns

Figure 1: 288 Groups and Companies

[154]:

Group	status	paks	music_show_awards	end_year_awards	physical_sales	organic_youtube_views	
4Minute	Inactive	0	23	7	205088	818287184	1
AOA	Inactive	0	10	3	223475	872310873	1
B.A.P	Inactive	0	8	4	911467	301681895	1
BIGBANG	Inactive	7	84	46	3514210	4589887632	1
Beast	Inactive	0	47	28	1087745	657013261	1
Block B	Inactive	0	8	1	418220	353939253	1
Busker Busker	Inactive	3	10	10	344002	44241494	1
CLC	Inactive	0	2	0	65774	420286429	1
Crayon Pop	Inactive	0	2	3	36845	188584801	1
GFriend	Inactive	1	71	13	999591	1145370316	1
Girl's Day	Inactive	0	12	4	127226	553105564	1
H.O.T	Inactive	0	0	0	0	16197860	1
I.O.I	Inactive	1	9	3	276039	297420417	1
IZ*ONE	Inactive	0	26	7	2101206	1051316370	1
LOONA	Inactive	0	4	1	698360	365528056	1
Miss A	Inactive	2	16	12	111748	502828620	1
Momoland	Inactive	0	11	4	66422	1881942223	1
NU'EST	Inactive	0	17	5	1176812	439100514	1
NewJeans	Inactive	2	35	32	8278782	2296351136	1
PENTAGON	Inactive	0	2	0	650777	733366148	1
SS501	Inactive	0	12	5	398819	225868285	1
Sistar	Inactive	2	48	25	137519	957775756	1
Trouble Maker	Inactive	0	12	2	66974	248786656	1
VICTON	Inactive	0	3	0	750171	180700492	1
WJSN	Inactive	0	9	1	850460	241820625	1
Wanna One	Inactive	0	49	21	3687191	578452979	1
Weeekly	Inactive	0	0	1	314973	318429943	1
Wonder Girls	Inactive	1	32	17	309310	586428007	1
f(x)	Inactive	0	33	10	548777	560004408	1

Name: count, dtype: int64

Figure 2: twenty-fourth percent of Group Inactive

Group	Company	followers	top_track_popularity	status	paks	music_show_awards	end_year_awards	physical_sales	Youtube_view_count
f(x)	SM	0.000000	0.000000	Inactive	0.0	0.018043	0.010515	0.001798	0.004458
EXO	SM	0.039178	0.036308	Active	0.0	0.062329	0.077813	0.044772	0.034409
NCT	SM	0.027920	0.028871	Active	0.0	0.021870	0.016824	0.046155	0.014526
Super Junior	SM	0.000000	0.000000	Active	0.0	0.028978	0.051525	0.020814	0.015861
aespa	SM	0.013805	0.036745	Active	2.0	0.019683	0.042061	0.028476	0.020428
Super Junior-M	SM	0.000000	0.000000	0	0.0	0.000000	0.000000	0.000000	0.000000
Red Velvet	SM	0.034624	0.034121	Active	1.0	0.046473	0.019979	0.017242	0.027034
SHINee	SM	0.012277	0.033246	Active	0.0	0.037726	0.035752	0.010240	0.010418

After cleaning the data, I proceeded with graph modeling to apply the second step: the PageRank algorithm. When performing graph modeling, it was necessary to carefully determine the weights in order to effectively apply PageRank. First, the columns designated for weighting were converted into proportions of their respective total sums, ensuring that the sum of the weights amounted to 1. For instance, given that HYBE's 2024 revenue was $1,650million, SM'swas$660 million, JYP's was $450million, andYG'swas$300 million, these figures were converted into the following proportions: (0.539, 0.216, 0.147, 0.098).

o	Company	followers	top_track_popularity	status	paks	music_show_awards	end_year_awards	physical_sales	Youtube_view_count	revenue
)	SM	0.000000	0.000000	Inactive	0.0	0.018043	0.010515	0.001798	0.004458	0.216
D	SM	0.039178	0.036308	Active	0.0	0.062329	0.077813	0.044772	0.034409	0.216
T	SM	0.027920	0.028871	Active	0.0	0.021870	0.016824	0.046155	0.014526	0.216
r r	SM	0.000000	0.000000	Active	0.0	0.028978	0.051525	0.020814	0.015861	0.216
a	SM	0.013805	0.036745	Active	2.0	0.019683	0.042061	0.028476	0.020428	0.216
r - M	SM	0.000000	0.000000	0	0.0	0.000000	0.000000	0.000000	0.000000	0.216
d t	SM	0.034624	0.034121	Active	1.0	0.046473	0.019979	0.017242	0.027034	0.216
e	SM	0.012277	0.033246	Active	0.0	0.037726	0.035752	0.010240	0.010418	0.216

Since using actual revenue figures here would likely result in HYBE being valued disproportionately higher, I have converted the data into ratios instead. I modeled the idol groups and companies as nodes, and utilized the other columns as weights.

The weights assigned to the group node's influence on its constituent nodes were calculated as follows: Music Awards * 0.4, End-Year Awards * 0.4, and Physical Sales * 0.2. The decision to assign significant weight to the frequency of Music Award and Year-End Award wins was based on the premise that entertainment agencies today primarily target the global market; furthermore, I reasoned that accumulating numerous awards facilitates global reach and helps cultivate a substantial fandom—factors that exert a profound influence on the agencies themselves.

Regarding the influence an agency exerts on a group, weights were assigned as follows: YouTube views * 0.6, follower count * 0.2, and recent popularity * 0.2. The rationale behind this is that YouTube is a widely popular platform already utilized by people worldwide, and an agency significantly impacts a group by actively managing and supporting its presence on the platform. While follower count and recent popularity are undoubtedly important from a marketing and promotional standpoint, they were assigned slightly lower weights due to certain gaps in the available data.

What was interesting here was the discovery that when the influence of the agency on an idol group was weighted based on actual revenue figures, it appeared to have almost no impact. Rather, the way major agencies influence their affiliated idol groups is by providing full-scale support for publicity and fandom management—a factor that significantly boosts their chances of success.

	node	rank
0	SM	0.030341
1	JYP	0.029680
2	YG	0.025728
3	HYBE	0.022580
4	Starship	0.007065
5	FNC	0.007065

Despite HYBE's revenue being so significantly higher than that of the other three agencies that they are hardly comparable, its PageRank ranking remains at fourth place.

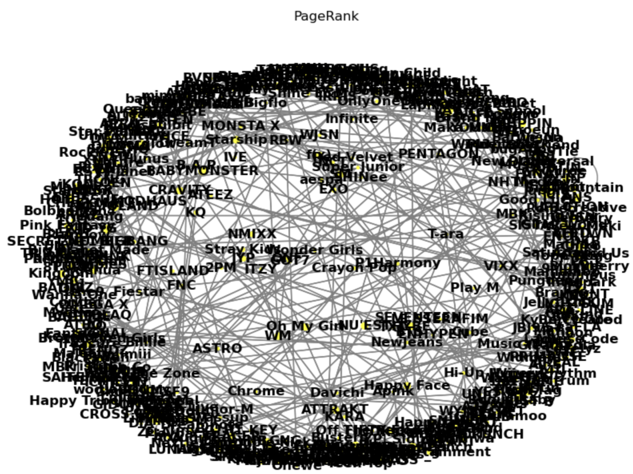
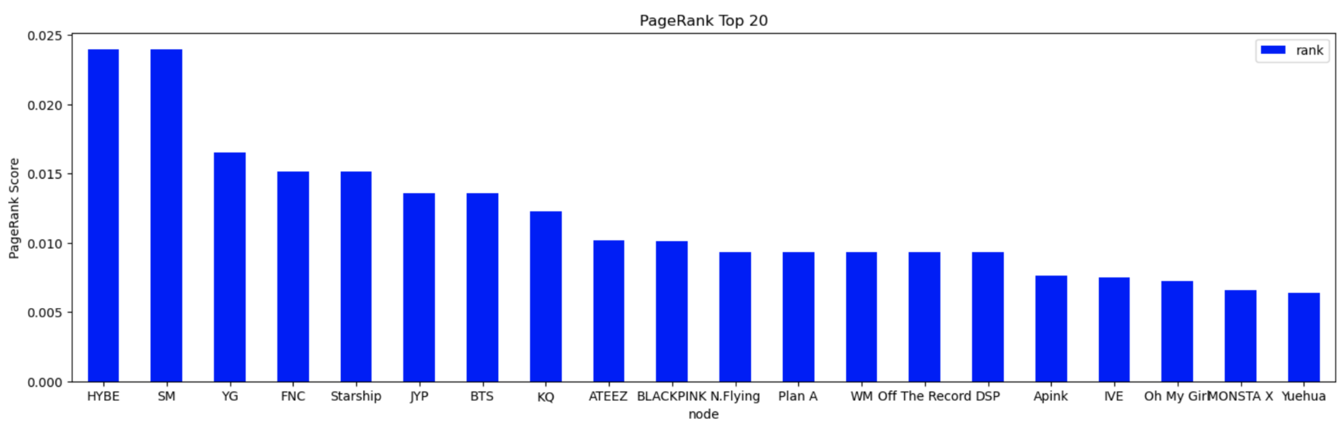


Figure 3: PageRank Graph Visualization

[183]:

	node	rank
0	HYBE	0.023968
1	SM	0.023968
2	YG	0.016538
3	FNC	0.015174
4	Starship	0.015174
5	JYP	0.013610
6	BTS	0.013551
7	KQ	0.012242
8	ATEEZ	0.010200
9	BLACKPINK	0.010140
10	N.Flying	0.009351
11	WM	0.009311
12	Plan A	0.009311
13	Off The Record	0.009311
14	DSP	0.009311
15	Apink	0.007625
16	IVE	0.007521
17	Oh My Girl	0.007234
18	MONSTA X	0.006601
19	Happy Face	0.006380

Figure 4: Top Twenty PageRank



Ultimately, the most influential agencies were HYBE and SM, and the most influential idol group is BTS.

During my research using these datasets, the most significant limitation I encountered was the scarcity of data-specific information. Data details were severely lacking, and as I utilized various datasets, I found that they often lacked consistency—specifically, the data points did not consistently originate from the same time period. Given the inadequacy of the existing datasets, I aspire to conduct extensive independent research in the future to curate and construct my own collection of useful datasets. Furthermore, assuming that fandoms play a crucial role, I intend to link influential fandoms directly to specific idol groups; by doing so, I hope to uncover more reliable empirical evidence regarding which follower metrics carry the most weight and to investigate the mechanisms through which this influence propagates. Furthermore, I would like to incorporate Billboard chart data and integrate information regarding the number of featured appearances, brand partnerships, and advertising collaborations to generate comprehensive results. Finally, I would like to collect individual data on the members of idol groups to investigate how their solo activities impact the group activities to which they belong.

Works Cited

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